



Tertiary Infotech Academy Pte Ltd
UEN: 201200696W

LESSON PLAN

For

AI Agents with Gemini Spark

TGS Ref No: TGS-2026064173

Conducted by
Tertiary Infotech Academy Pte Ltd
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Version 1.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	23 August 2026	Initial release - one-day aligned schedule with actual deck slide references and PP/OQ assessment.	Dr. Alfred Ang

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Course Information

Course title	AI Agents with Gemini Spark
Course code	TGS-2026064173
Training provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	8 hours, including assessment
Delivery	Instructor-led concepts, demonstrations, guided labs and evidence review
Assessment	Practical Performance 60 minutes + Oral Questioning 20 minutes
Trainer	Dr. Alfred Ang

Learning Outcomes

- LO1: Determine an AI-agent strategy by translating a web-design goal into bounded tasks, permissions, risks and human approval points.
- LO2: Develop web content and a responsive website through a controlled vibe-coding loop using Claude Code, Codex or Antigravity, with Gemini Spark supporting research and reusable skills.
- LO3: Develop an implementation plan for Gemini Spark tasks, skills, schedules and connected tools, including responsibilities, timelines and rollback controls.
- LO4: Evaluate agent and website performance using evidence, acceptance criteria, incident signals and iterative improvement decisions.

Daily Schedule

Time	Duration	Topic / Activity	Method	Slides
9:00-9:20	20 min	Welcome, digital attendance, course orientation and assessment briefing	Briefing / facilitation	2-15
9:20-10:20	60 min	Topic 1 - Agent Strategy and Gemini Spark Runtime	Concepts / demonstration	22-38
10:20-10:30	10 min	Tea break	Break	—
10:30-11:20	50 min	Lab 1 - Create the Agent Strategy and Approval Map	Guided practical	35
11:20-12:10	50 min	Topic 2 - Vibe Coding a Responsive Website	Concepts / demonstration	39-55
12:10-12:50	40 min	Lunch break	Lunch	—
12:50-13:40	50 min	Lab 2 - Vibe Code the Northstar Studio Website	Guided practical	52
13:40-14:30	50 min	Topic 3 - Skills, Schedules and Implementation Governance	Concepts / demonstration	56-72
14:30-14:40	10 min	Tea break	Break	—
14:40-15:30	50 min	Lab 3 - Design a Reusable Spark Content Skill and Schedule	Guided practical	69
15:30-16:10	40 min	Topic 4 - Monitoring, Incident Control and Optimisation	Concepts / demonstration	73-89
16:10-16:30	20 min	Lab 4 - Monitor, Contain and Optimise an Agent Run	Guided practical	86

Time	Duration	Topic / Activity	Method	Slides
16:30-16:40	10 min	Final assessment briefing and assessment digital attendance	Briefing / facilitation	94-96
16:40-17:40	60 min	Practical Performance (PP)	Summative assessment	—
17:40-18:00	20 min	Oral Questioning (OQ) and sign-off	Summative assessment	—

Instructional time: 400 minutes · Assessment: 80 minutes · Breaks: 20 minutes · Lunch: 40 minutes.

Topic-by-Topic Facilitation

Topic 01 - Agent Strategy and Gemini Spark Runtime • Slides 22-38

Outcome and criteria: LO1 · K1, A1

Facilitation focus: Autonomy is safe only when the goal, permission boundary, evidence and stop conditions are explicit.

- Task thread: Holds the goal, instructions, files and task state.
- Planner: Decomposes the goal into observable steps and revises when blocked.
- Skills: Reusable how-to instructions and context, selected explicitly or automatically.
- Connected tools: Workspace apps, websites, local or remote browser, and code execution.
- Human control: Review, confirm, take over, stop, pause schedules and inspect changed files.

Lab 1 - Create the Agent Strategy and Approval Map • Slide 35

Build: A strategy brief, task contract, permission matrix, risk register and human-approval map for the Northstar Studio website.

Acceptance: The task can be executed without guessing: every step has an owner, allowed tools, a risk tier, approval rule, stop condition and observable completion evidence.

Trainer evidence check: Review the requested artifact, test result, permission boundary and approval record before moving on.

Topic 02 - Vibe Coding a Responsive Website • Slides 39-55

Outcome and criteria: LO2 · K4, K5, A2

Facilitation focus: Vibe coding is rapid co-creation with an AI coding agent, disciplined by small changes, executable evidence and human review.

- Design brief: Audience, job-to-be-done, content, constraints and acceptance tests.
- Coding agent: Claude Code, Codex or Antigravity edits the repository from natural-language intent.
- Local project: HTML, CSS, JavaScript, assets, README and version history.
- Test loop: Open in a browser, check responsive layouts, links, form states and accessibility.
- Reviewer: Accepts, rejects or redirects each meaningful change before release.

Lab 2 - Vibe Code the Northstar Studio Website • Slide 52

Build: A working one-page HTML/CSS/JavaScript website, responsive at three widths, with accessible navigation, an evidence log and a rollback commit.

Acceptance: The website renders without errors, passes the supplied structural checks, works at desktop/tablet/mobile widths, is keyboard navigable, contains only approved claims, and has an identifiable rollback commit.

Trainer evidence check: Review the requested artifact, test result, permission boundary and approval record before moving on.

Topic 03 - Skills, Schedules and Implementation Governance • Slides 56-72

Outcome and criteria: LO3 · K3, K5, A3

Facilitation focus: A reusable agent workflow is an operating procedure with versioned instructions, scoped tools, owners and measurable completion evidence.

- Task: The what: a complete project or objective managed in a thread.
- Skill: The how: reusable instructions, context, tools and quality rules.
- Schedule: The when: a time-based or event-based trigger for a task.
- Connected apps: The where: Gmail, Calendar, Drive, Docs, Sheets, Slides, Search and approved third parties.
- Runbook: Owner, approvals, logs, exception handling, rollback and review cadence.

Lab 3 - Design a Reusable Spark Content Skill and Schedule • Slide 69

Build: A versioned website-content skill, output schema, schedule specification, trigger deduplication rule and implementation plan.

Acceptance: The skill produces the specified draft and source log, the schedule cannot overlap, publication is impossible without human approval, and every stage has an owner, target time and evidence artifact.

Trainer evidence check: Review the requested artifact, test result, permission boundary and approval record before moving on.

Topic 04 - Monitoring, Incident Control and Optimisation • Slides 73-89

Outcome and criteria: LO4 · K2, K6, A4

Facilitation focus: An agent is not successful because it completed a run; it is successful when the output is correct, safe, useful and recoverable.

- Progress panel: Planned, current and completed steps reveal runtime state.
- Files panel: Shows which files were read or updated and supports evidence review.
- Schedules panel: Pause or resume recurring work and inspect the active trigger.
- Takeover/Stop: Human interrupts browsing, supplies sensitive data directly, or cancels the task.
- Scorecard: Quality, safety, efficiency and business-result indicators drive the next change.

Lab 4 - Monitor, Contain and Optimise an Agent Run • Slide 86

Build: A completed KPI scorecard, incident timeline, containment record, root-cause analysis and verified improvement proposal.

Acceptance: The learner detects the unsafe instruction, stops and contains the run, restores the approved state, explains the root cause, and proves one control improvement with before/after evidence.

Trainer evidence check: Review the requested artifact, test result, permission boundary and approval record before moving on.

Resources Required

- Trainer slide deck and projector
- Learner Guide and four self-contained lab folders
- Modern browser and a code editor/terminal

- Claude Code, Codex or Antigravity for Lab 2
- Gemini Spark access where available; supplied offline simulation where it is not
- Git and Python 3 for the website lab checks
- LMS access for materials, attendance and assessment

Assessment

- Practical Performance (PP) - 4 tasks covering A1-A4 - 60 minutes - open book.
- Oral Questioning (OQ) - 4 questions covering K1-K6 - 20 minutes - open book.
- All criteria must be demonstrated as competent. Candidate papers are submitted on the LMS; answer keys remain trainer-only.
- Assessment briefing begins at 16:30; PP runs 16:40-17:40; OQ and sign-off run 17:40-18:00.

Assessment portal: <https://lms-tms.tertiaryinfotech.com/>